

```
#include <stdbool.h>
```

```
#include <stdio.h>
```

```
#define MAX 5
```

```
int QUEUE[MAX], REAR = -1, FRONT = -1;
```

```
void insertRear();
```

```
void deleteFront();
```

```
void deleteRear();
```

```
void insertFront();
```

```
void display();
```

```
bool isFull();
```

```
bool isEmpty();
```

```
int main() {
```

```
    int choice;
```

```
    printf("Menu:\n 1. Insert from Rear\n 2. Delete from Front\n 3. "
```

```
        "Insert from Front\n 4. Delete from Rear\n 5. Display\n 6. Exit\n");
```

```
    while (true) {
```

```
        printf("\nEnter choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice) {
```

```
case 1:
```

```
    insertRear();
```

```
    break;
```

```
case 2:
```

```
    deleteFront();
```

```
    break;
```

```
case 3:
```

```
    insertFront();
```

```
    break;
```

```
case 4:
```

```
    deleteRear();
```

```
    break;
```

```
case 5:
```

```
    display();
```

```
    break;
```

```
case 6:
```

```
    printf("Exiting...\n");
```

```
    return 0;
```

```
default:
```

```
    printf("Invalid choice");
```

```
}
```

```
}
```

```
return 0;
```

```
}
```

```
bool isFull() { return REAR == MAX - 1; }
```

```
bool isEmpty() { return FRONT < 0; }
```

```
void insertRear() {
```

```
    int a;
```

```
    if (isFull()) {
```

```
        printf("Queue is full");
```

```
        return;
```

```
    }
```

```
    if (FRONT == -1)
```

```
        FRONT++;
```

```
    printf("Enter Element: ");
```

```
    scanf("%d", &a);
```

```
    QUEUE[++REAR] = a;
```

```
}
```

```
void deleteFront() {
```

```
    int value;
```

```
    if (isEmpty()) {
```

```
        printf("Queue Empty");
```

```
        return;
```

```
    }
```

```
    value = QUEUE[FRONT];
```

```
    if (REAR == FRONT)
```

```
        FRONT = REAR = -1;
```

```
    else
```

```
        FRONT++;
```

```
    printf("The deleted element from front is: %d", value);
```

```
}
```

```
void deleteRear() {
```

```
    int value;
```

```
if (isEmpty()) {  
    printf("Queue Empty");  
    return;  
}
```

```
value = QUEUE[REAR];
```

```
if (REAR == FRONT)  
    FRONT = REAR = -1;  
else  
    REAR--;
```

```
printf("The deleted element from rear is: %d", value);  
}
```

```
void insertFront() {
```

```
    if (FRONT == 0) {  
        printf("Front is at the beginning - insertion not possible");  
        return;  
    }
```

```
    if (FRONT == -1) {  
        FRONT++;  
        REAR++;
```

```
} else
```

```
    FRONT--;
```

```
int x;
```

```
printf("Enter Element: ");
```

```
scanf("%d", &x);
```

```
    QUEUE[FRONT] = x;
```

```
}
```

```
void display() {
```

```
    if (isEmpty()) {
```

```
        printf("Queue Empty");
```

```
        return;
```

```
    }
```

```
    printf("Printing DeQueue:\n");
```

```
    for (int i = FRONT; i <= REAR; i++)
```

```
        printf("\t%d", QUEUE[i]);
```

```
}
```